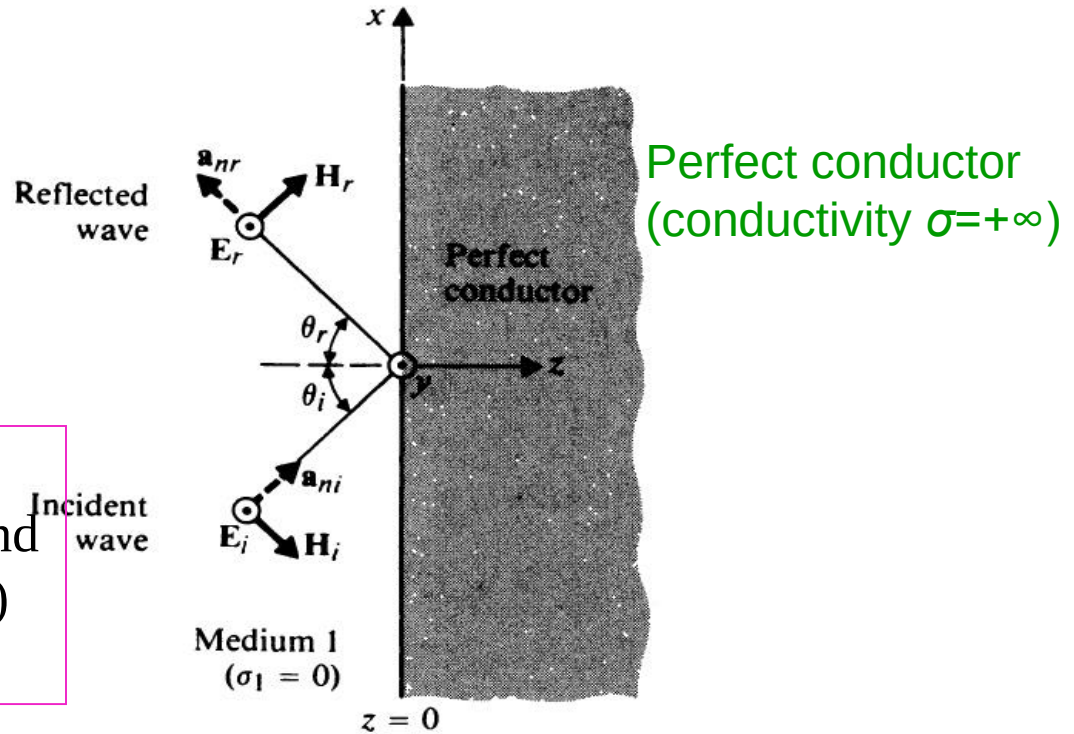


§ 8-7 Oblique Incidence at a Plane Conducting Boundary

Lossless medium
(conductivity $\sigma=0$)

Incidence plane(入射面):

The incident wave vector $\mathbf{k}$ and the normal $\mathbf{n}$ (of the interface) form the incidence plane



- 1) **Perpendicular polarization (TE or s-polarization):** $\mathbf{E}$ is perpendicular to the plane of incidence
- 2) **Parallel polarization (TM, p-polarization):** $\mathbf{E}$ is parallel to the plane of incidence

8-7.1 Perpendicular polarization (垂直偏振)

Electric field $\mathbf{E}_1$ in medium 1:

Incident direction $\mathbf{a}_{ni}$: $\mathbf{a}_{ni} = \mathbf{a}_x \sin \theta_i + \mathbf{a}_z \cos \theta_i$

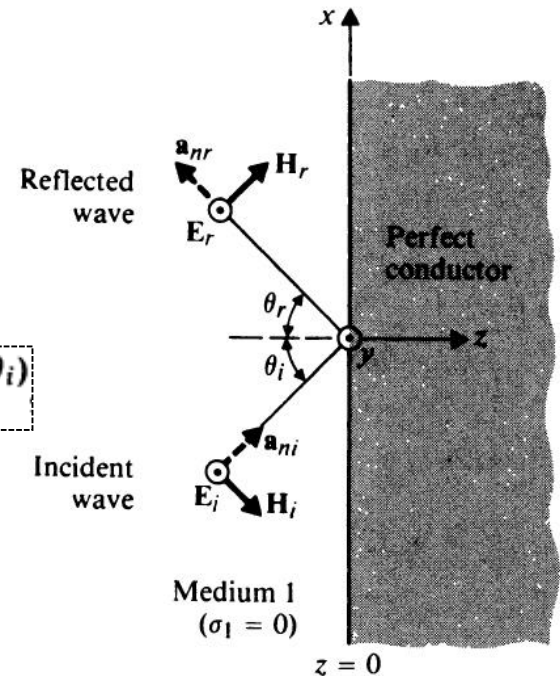
Incident field $\mathbf{E}_i$: (θ_i -incident angle)

$$\mathbf{E}_i(x, z) = \mathbf{a}_y E_{i0} e^{-j\beta_1 \mathbf{a}_{ni} \cdot \mathbf{R}} = \mathbf{a}_y E_{i0} e^{-j\beta_1 (x \sin \theta_i + z \cos \theta_i)}$$

Reflection direction $\mathbf{a}_{nr}$: $\mathbf{a}_{nr} = \mathbf{a}_x \sin \theta_r - \mathbf{a}_z \cos \theta_r$

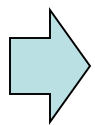
Reflection field $\mathbf{E}_r$: (θ_r -reflection angle)

$$\mathbf{E}_r(x, z) = \mathbf{a}_y E_{r0} e^{-j\beta_1 (x \sin \theta_r - z \cos \theta_r)}$$

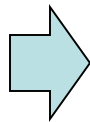


boundary condition, $\mathbf{E}_1(0) = \mathbf{E}_2(0)$

$$\mathbf{E}_1(x, 0) = \mathbf{E}_i(x, 0) + \mathbf{E}_r(x, 0) = \mathbf{a}_y (E_{i0} e^{-j\beta_1 x \sin \theta_i} + E_{r0} e^{-j\beta_1 x \sin \theta_r}) = 0.$$



$$E_{r0} = -E_{i0}$$



$$\mathbf{E}_r(x, z) = -\mathbf{a}_y E_{i0} e^{-j\beta_1 (x \sin \theta_i - z \cos \theta_i)}$$

$$\theta_r = \theta_i$$

(反射定律: Snell's law of reflection)

Total electric field $\mathbf{E}_1$:

$$\begin{aligned}\mathbf{E}_1(x, z) &= \mathbf{E}_i(x, z) + \mathbf{E}_r(x, z) \\ &= \mathbf{a}_y E_{i0} (e^{-j\beta_1 z \cos \theta_i} - e^{j\beta_1 z \cos \theta_i}) e^{-j\beta_1 x \sin \theta_i} \\ &= -\mathbf{a}_y j 2 E_{i0} \sin(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}.\end{aligned}$$

Magnetic field $\mathbf{E}_1$ in medium 1:

Incident field $\mathbf{H}_i$:

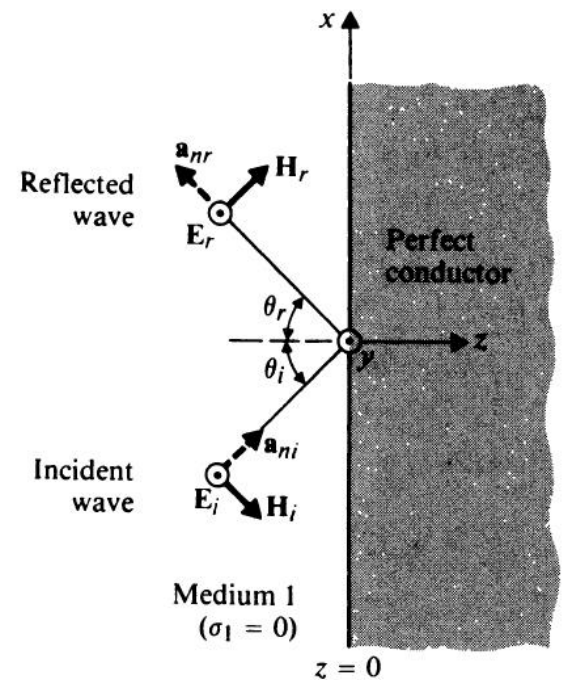
$$\mathbf{H}_i(x, z) = \frac{1}{\eta_1} [\mathbf{a}_{ni} \times \mathbf{E}_i(x, z)] = \frac{E_{i0}}{\eta_1} (-\mathbf{a}_x \cos \theta_i + \mathbf{a}_z \sin \theta_i) e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}.$$

Reflection field $\mathbf{H}_r$:

$$\mathbf{H}_r(x, z) = \frac{1}{\eta_1} [\mathbf{a}_{nr} \times \mathbf{E}_r(x, z)] = \frac{E_{i0}}{\eta_1} (-\mathbf{a}_x \cos \theta_i - \mathbf{a}_z \sin \theta_i) e^{-j\beta_1(x \sin \theta_i - z \cos \theta_i)}.$$

Total magnetic field $\mathbf{H}_1$:

$$\begin{aligned}\mathbf{H}_1(x, z) &= -2 \frac{E_{i0}}{\eta_1} [\mathbf{a}_x \cos \theta_i \cos(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i} \\ &\quad + \mathbf{a}_z j \sin \theta_i \sin(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}].\end{aligned}$$



DISCUSSIONS: (Oblique incidence of perpendicular polarization)

1. In the direction (z-direction) normal to the boundary, E_{1y} and H_{1x} maintain standing-wave patterns according to $\sin \beta_{1z}z$ and $\cos \beta_{1z}z$, respectively, where $\beta_{1z} = \beta_1 \cos \theta_i$. No average power is propagated in this direction since E_{1y} and H_{1x} are 90° out of time phase.
2. In the direction (x-direction) parallel to the boundary, E_{1y} and H_{1z} are in both time and space phase and propagate with a phase velocity

$$u_{1x} = \frac{\omega}{\beta_{1x}} = \frac{\omega}{\beta_1 \sin \theta_i} = \frac{u_1}{\sin \theta_i}. \quad (8-115)$$

The wavelength in this direction is

$$\lambda_{1x} = \frac{2\pi}{\beta_{1x}} = \frac{\lambda_1}{\sin \theta_i}. \quad (8-116)$$

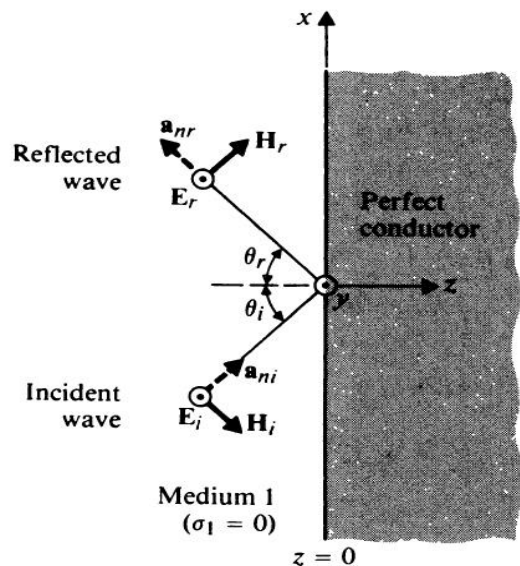


Fig. 8-11 (perpendicular polarization)
E is perpendicular to the plane of incidence.

$$\mathbf{E}_1(x, z) = -\mathbf{a}_y j 2 E_{i0} \sin(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}.$$

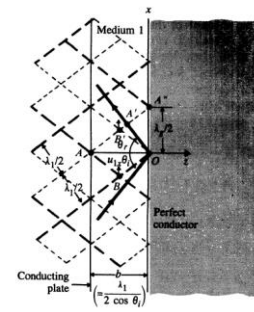
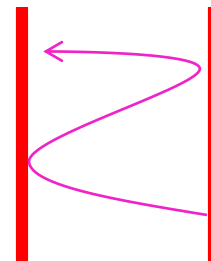
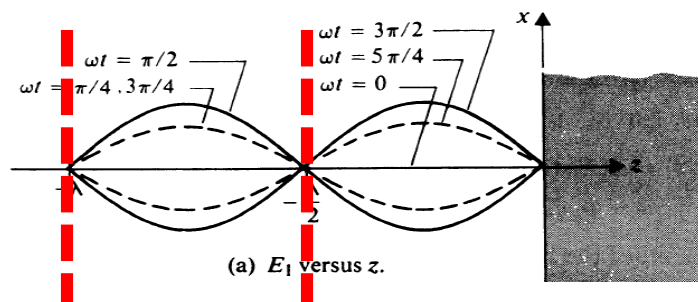
3. The propagating wave in the x direction is a **nonuniform plane wave** because its amplitude varies with z .
4. Since $\mathbf{E}_1 = 0$ for all x when $\sin(\beta_1 z \cos \theta_i) = 0$ or when

$$\beta_1 z \cos \theta_i = \frac{2\pi}{\lambda_1} z \cos \theta_i = -m\pi, \quad m = 1, 2, 3, \dots,$$

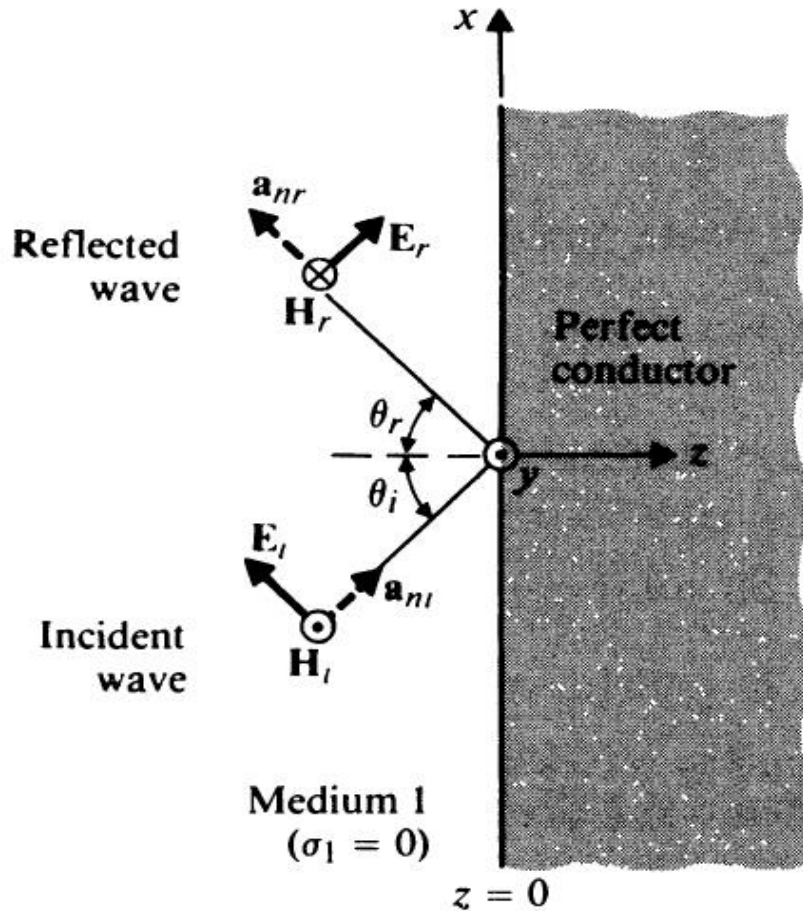
a conducting plate could be inserted at

$$z = -\frac{m\lambda_1}{2 \cos \theta_i}, \quad m = 1, 2, 3, \dots, \quad (8-117)$$

without changing the field pattern that exists between the conducting plate and the conducting boundary at $z = 0$. A **transverse electric (TE) wave** ($E_{1x} = 0$) would bounce back and forth between the conducting planes and propagate in the x -direction. We would have, in effect, a parallel-plate waveguide.



§ 8-7.2 Parallel polarization (平行极化)



E-polarization is in the incident plane; **H** is perpendicular to it.

Incident direction $\mathbf{a}_{ni}$:

$$\mathbf{a}_{ni} = \mathbf{a}_x \sin \theta_i + \mathbf{a}_z \cos \theta_i$$

Reflection direction $\mathbf{a}_{nr}$:

$$\mathbf{a}_{nr} = \mathbf{a}_x \sin \theta_r - \mathbf{a}_z \cos \theta_r$$

Incident **E** and **H** fields:

$$\mathbf{E}_i(x, z) = E_{i0}(\mathbf{a}_x \cos \theta_i - \mathbf{a}_z \sin \theta_i)e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)},$$

$$\mathbf{H}_i(x, z) = \mathbf{a}_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}.$$

Reflected **E** and **H** fields:

$$\mathbf{E}_r(x, z) = E_{r0}(\mathbf{a}_x \cos \theta_r + \mathbf{a}_z \sin \theta_r)e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)},$$

$$\mathbf{H}_r(x, z) = -\mathbf{a}_y \frac{E_{r0}}{\eta_1} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)}.$$

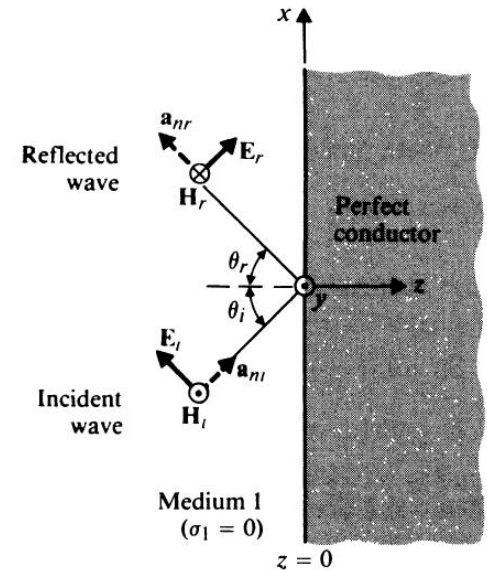
Electric field $\mathbf{E}_1$ in medium 1:

Incident field $\mathbf{E}_i$:

$$\mathbf{E}_i(x, z) = E_{i0}(\mathbf{a}_x \cos \theta_i - \mathbf{a}_z \sin \theta_i)e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}$$

Reflected field $\mathbf{E}_r$:

$$\mathbf{E}_r(x, z) = E_{r0}(\mathbf{a}_x \cos \theta_r + \mathbf{a}_z \sin \theta_r)e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)},$$



At $z=0$, according to the boundary condition, $\mathbf{E}_x(0) = 0$, we have

$$(E_{i0} \cos \theta_i)e^{-j\beta_1 x \sin \theta_i} + (E_{r0} \cos \theta_r)e^{-j\beta_1 x \sin \theta_r} = 0, \Rightarrow E_{r0} = -E_{i0} \text{ and } \theta_r = \theta_i.$$

Total electric field $\mathbf{E}_1$:

$$\begin{aligned} \mathbf{E}_1(x, z) &= \mathbf{E}_i(x, z) + \mathbf{E}_r(x, z) \\ &= -2E_{i0}[\mathbf{a}_x j \cos \theta_i \sin (\beta_1 z \cos \theta_i) \\ &\quad + \mathbf{a}_z \sin \theta_i \cos (\beta_1 z \cos \theta_i)]e^{-j\beta_1 x \sin \theta_i}. \end{aligned}$$

Total magnetic field $\mathbf{H}_1$:

$$\mathbf{H}_1(x, z) = \mathbf{H}_i(x, z) + \mathbf{H}_r(x, z)$$

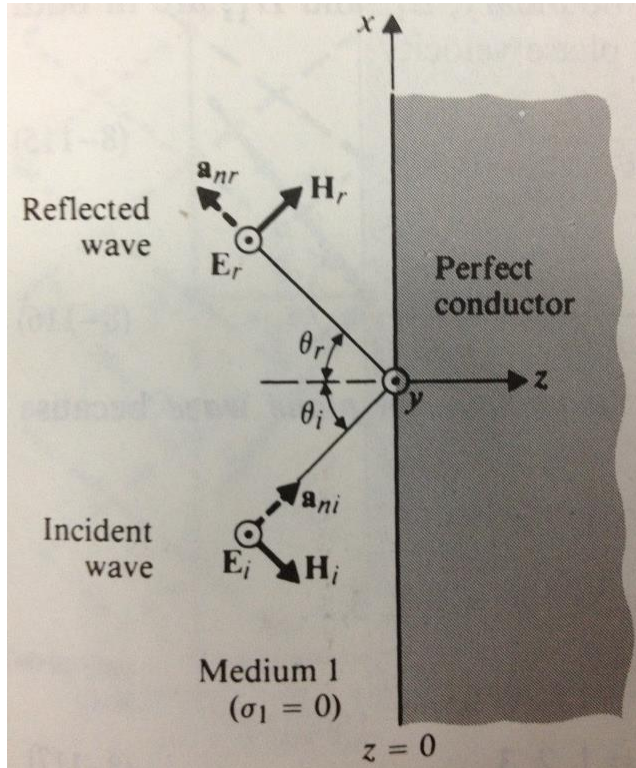
$$= \mathbf{a}_y 2 \frac{E_{i0}}{\eta_1} \cos (\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}.$$

DISCUSSIONS: (Oblique incidence of parallel polarization)

$$\mathbf{H}_1(x, z) = \mathbf{a}_y 2 \frac{E_{i0}}{\eta_1} \cos(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}.$$

1. In the direction (z-direction) normal to the boundary, E_{1x} and H_{1y} maintain standing-wave patterns according to $\sin \beta_{1z} z$ and $\cos \beta_{1z} z$, respectively, where $\beta_{1z} = \beta_1 \cos \theta_i$. No average power is propagated in this direction, since E_{1x} and H_{1y} are 90° out of time phase.
2. In the x-direction parallel to the boundary, E_{1z} and H_{1y} are in both time and space phase and propagate with a phase velocity $u_{1x} = u_1 / \sin \theta_i$, which is the same as that in the perpendicular polarization.
3. As in the case of perpendicular polarization, the propagating wave in the x-direction is a nonuniform plane wave.
4. The insertion of a conducting plate at $z = -m\lambda_1/2 \cos \theta_i$ ($m = 1, 2, 3, \dots$) where $E_{1x} = 0$ for all x would not affect the field pattern that exists between the conducting plate and the conducting boundary at $z = 0$; we would then have a parallel-plate waveguide. A **transverse magnetic (TM) wave** ($H_{1x} = 0$) will propagate in the x-direction. (TM waves between parallel plates will be discussed in Subsection 10–3.1.)

8-10 A uniform plane wave $(\vec{E}_i, \vec{H}_i)$ of a angular frequency ω is incident from air on a very large, perfectly conducting wall at an angle of incidence θ_i with perpendicular polarization. Find (a) the current induced on the wall surface, and (b) the time-average Poynting vector in medium 1.



$$\mathbf{H}_1(x, z) = -2 \frac{E_{i0}}{\eta_1} [\mathbf{a}_x \cos \theta_i \cos (\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i} + \mathbf{a}_z j \sin \theta_i \sin (\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}].$$

$$\vec{H}_1(x, 0) = -\frac{E_{i0}}{\eta_0} (\vec{a}_x 2 \cos \theta_i) e^{-j\beta_0 x \sin \theta_i}.$$

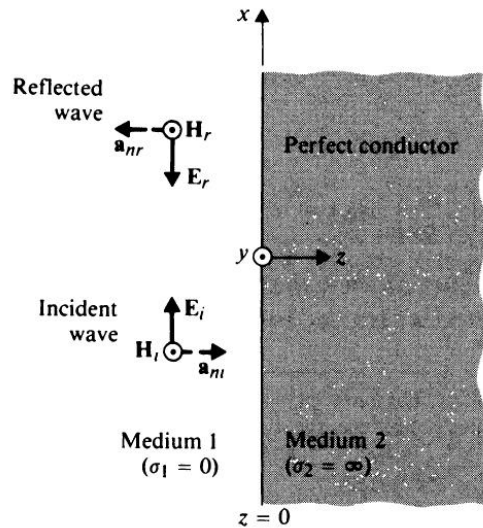
Inside the perfectly conducting wall, both $\mathbf{E}_2$ and $\mathbf{H}_2$ must vanish. There is then a discontinuity in the magnetic field.

The amount of discontinuity is equal to the surface current.

$$\vec{J}_s(x) = \vec{a}_{n2} \times \vec{H}_1(x, 0) = (-\vec{a}_z) \times (-\vec{a}_x) \frac{E_{i0}}{\eta_0} (2 \cos \theta_i) e^{-j\beta_0 x \sin \theta_i} = \vec{a}_y \frac{E_{i0}}{60\pi} (\cos \theta_i) e^{-j(\omega/c)x \sin \theta_i}.$$

Revisit: Incidence at a Plane Conducting Boundary

Normal Incidence

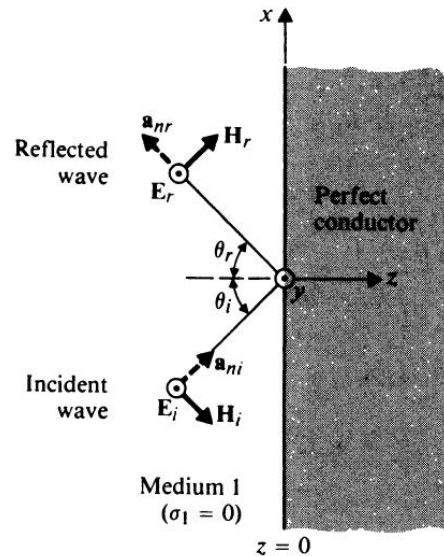


$$\mathbf{E}_i(z) = \mathbf{a}_x E_{i0} e^{-j\beta_1 z}$$

$$\mathbf{E}_{r0} = -\mathbf{E}_{i0}$$

$$\begin{aligned} \mathbf{E}_1(z) &= \mathbf{a}_x E_{i0} (e^{-j\beta_1 z} - e^{+j\beta_1 z}) \\ &= -\mathbf{a}_x j 2 E_{i0} \sin \beta_1 z. \end{aligned}$$

Perpendicular polarization

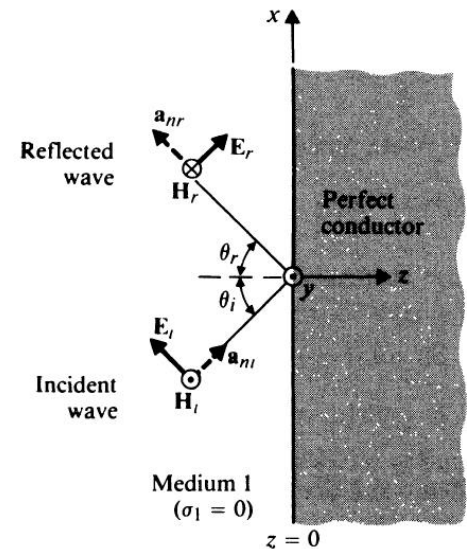


$$\mathbf{E}_i(x, z) = \mathbf{a}_y E_{i0} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}$$

$$\mathbf{E}_r(x, z) = -\mathbf{a}_y E_{i0} e^{-j\beta_1(x \sin \theta_i - z \cos \theta_i)}$$

$$\begin{aligned} \mathbf{E}_1(x, z) &= \mathbf{E}_i(x, z) + \mathbf{E}_r(x, z) \\ &= \mathbf{a}_y E_{i0} (e^{-j\beta_1 z \cos \theta_i} - e^{j\beta_1 z \cos \theta_i}) e^{-j\beta_1 x \sin \theta_i} \\ &= -\mathbf{a}_y j 2 E_{i0} \sin(\beta_1 z \cos \theta_i) e^{-j\beta_1 x \sin \theta_i}. \end{aligned}$$

Parallel polarization



$$\mathbf{E}_i(x, z) = E_{i0} (\mathbf{a}_x \cos \theta_i - \mathbf{a}_z \sin \theta_i) e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}$$

$$\mathbf{E}_r(x, z) = -E_{i0} (\mathbf{a}_x \cos \theta_i + \mathbf{a}_z \sin \theta_i) e^{-j\beta_1(x \sin \theta_i - z \cos \theta_i)}$$

$$\begin{aligned} \mathbf{E}_1(x, z) &= \mathbf{E}_i(x, z) + \mathbf{E}_r(x, z) \\ &= -2 E_{i0} [\mathbf{a}_x j \cos \theta_i \sin(\beta_1 z \cos \theta_i) \\ &\quad + \mathbf{a}_z \sin \theta_i \cos(\beta_1 z \cos \theta_i)] e^{-j\beta_1 x \sin \theta_i}. \end{aligned}$$

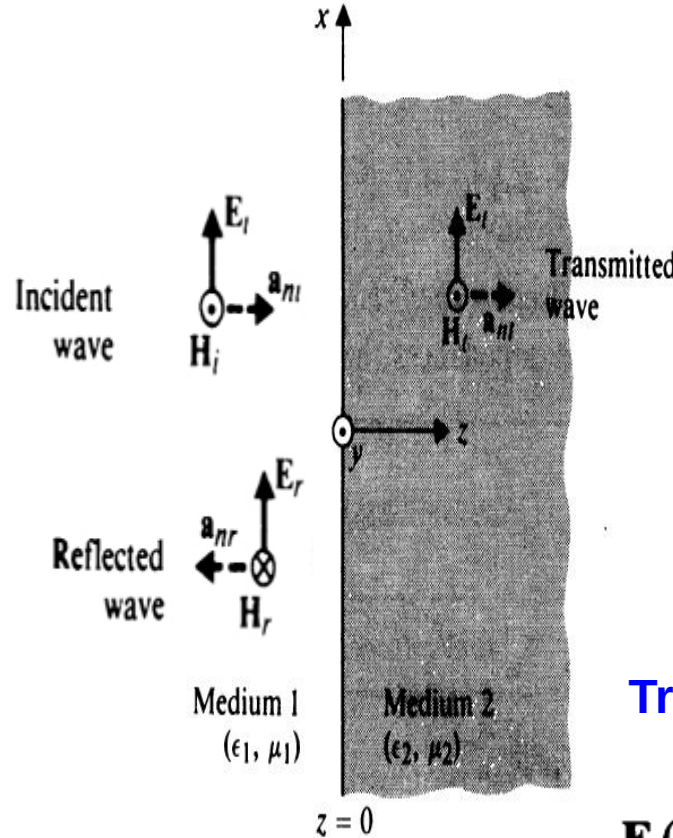
§ 8-8 Normal Incidence at a Plane Dielectric Boundary

Both media are dissipationless (electric conductivity $\sigma=0$ $j=0$)

Incident fields:

$$\mathbf{E}_i(z) = \mathbf{a}_x E_{i0} e^{-j\beta_1 z},$$

$$\mathbf{H}_i(z) = \mathbf{a}_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1 z}.$$



Reflected fields:

$$\mathbf{E}_r(z) = \mathbf{a}_x E_{r0} e^{j\beta_1 z},$$

$$\mathbf{H}_r(z) = (-\mathbf{a}_z) \times \frac{1}{\eta_1} \mathbf{E}_r(z) = -\mathbf{a}_y \frac{E_{r0}}{\eta_1} e^{j\beta_1 z}.$$

Boundary conditions ($z=0$)

$$E_{1t} = E_{2t}, \quad H_{1t} = H_{2t}$$



Find E_{r0} and E_{t0}

Transmitted (透射) fields:

$$\mathbf{E}_t(z) = \mathbf{a}_x E_{t0} e^{-j\beta_2 z},$$

$$\mathbf{H}_t(z) = \mathbf{a}_z \times \frac{1}{\eta_2} \mathbf{E}_t(z) = \mathbf{a}_y \frac{E_{t0}}{\eta_2} e^{-j\beta_2 z},$$

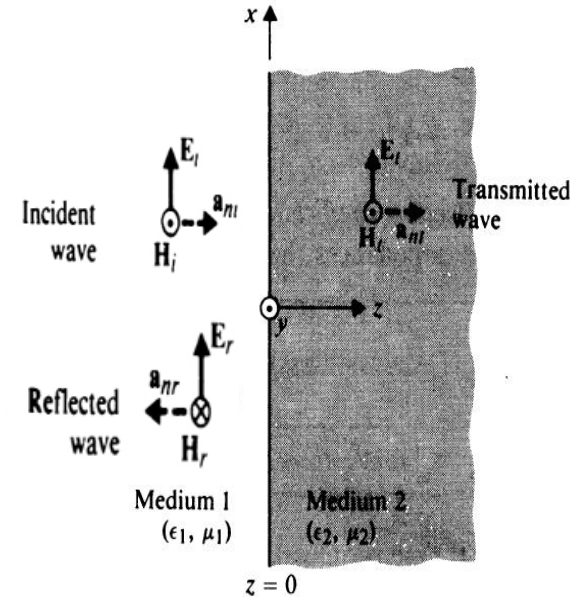
Reflection and transmission coefficients (反射与透射系数)

boundary conditions at $z=0$

$$\mathbf{E}_i(0) + \mathbf{E}_r(0) = \mathbf{E}_t(0) \quad \text{or} \quad E_{i0} + E_{r0} = E_{t0}$$

$$\mathbf{H}_i(0) + \mathbf{H}_r(0) = \mathbf{H}_t(0) \quad \text{or} \quad \frac{1}{\eta_1} (E_{i0} - E_{r0}) = \frac{E_{t0}}{\eta_2}$$

$$\eta = \frac{\omega\mu}{k} = \sqrt{\frac{\mu}{\epsilon}} \quad (\Omega)$$



$$E_{r0} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} E_{i0},$$

$$E_{t0} = \frac{2\eta_2}{\eta_2 + \eta_1} E_{i0}.$$

Reflection coefficient: $\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$

Transmission coefficient: $\tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1}$

$$1 + \Gamma = \tau$$

Standing wave ratio (驻波比)

Reflection coefficient:

$$\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

□ If medium 2 is a PEC ($\eta=0$):

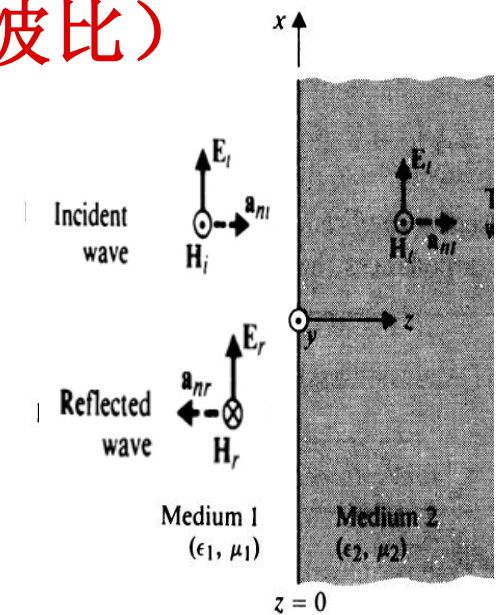
$$\Gamma = -1, \tau = 0$$

□ If medium 2 is not a good conductor,

$$\begin{aligned} \mathbf{E}_1(z) &= \mathbf{E}_i(z) + \mathbf{E}_r(z) = \mathbf{a}_x E_{i0} (e^{-j\beta_1 z} + \Gamma e^{j\beta_1 z}) \\ &= \mathbf{a}_x E_{i0} [(1 + \Gamma) e^{-j\beta_1 z} + \Gamma (e^{j\beta_1 z} - e^{-j\beta_1 z})] \\ &= \mathbf{a}_x E_{i0} [\underbrace{(1 + \Gamma) e^{-j\beta_1 z}}_{\text{Travelling wave}} + \underbrace{\Gamma (j2 \sin \beta_1 z)}_{\text{Standing wave}}] \end{aligned}$$

Travelling
wave

Standing
wave



$$\eta_c = \sqrt{\frac{\mu}{\epsilon_r}} \approx \sqrt{\frac{j\omega\mu}{\sigma}}$$

§ 8-10 Oblique Incidence at a Plane Dielectric Boundary

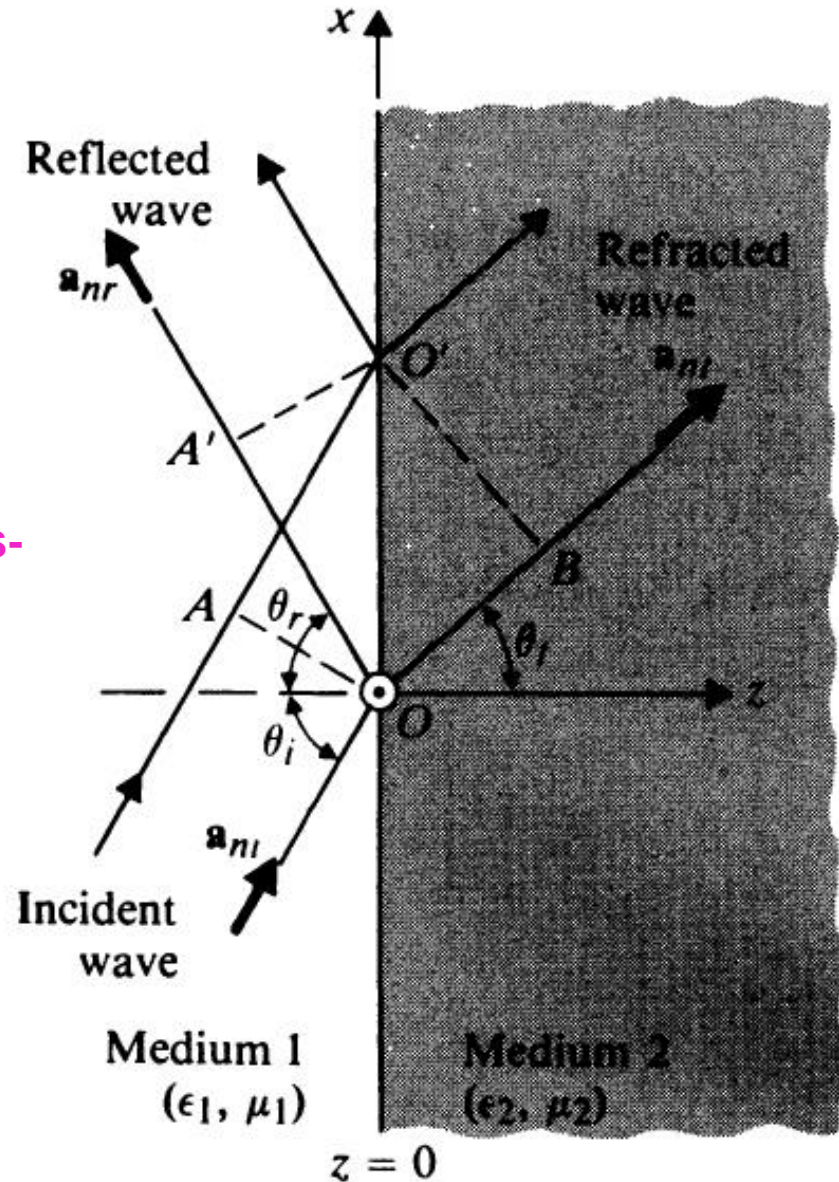
Both media are dissipationless
(electric conductivity $\sigma=0$)

(1) Perpendicular polarization (TE, s-polarization);

$$E_y, H_x, H_z$$

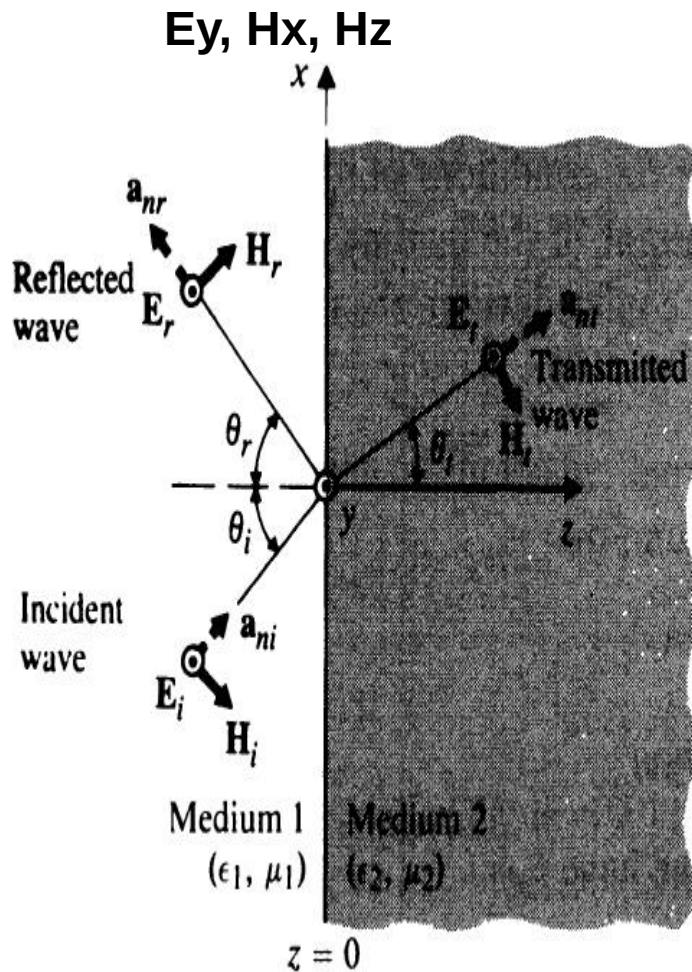
(2) Parallel polarization (TM, p-polarization);

$$E_x, E_z, H_y$$



§ 8-10.2 Perpendicular polarization (TE, s-polarization)

Perpendicular polarization (**E** is perpendicular to the plane of incidence)



Utilizing boundary conditions to solve E_{r0} , θ_r , E_{t0} , θ_t

Incident fields:

$$\mathbf{E}_i(x, z) = \mathbf{a}_y E_{i0} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}$$

$$\mathbf{H}_i(x, z) = \frac{E_{i0}}{\eta_1} (-\mathbf{a}_x \cos \theta_i + \mathbf{a}_z \sin \theta_i) e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}$$

Reflected fields:

$$\mathbf{E}_r(x, z) = \mathbf{a}_y E_{r0} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)}$$

$$\mathbf{H}_r(x, z) = \frac{E_{r0}}{\eta_1} (\mathbf{a}_x \cos \theta_r + \mathbf{a}_z \sin \theta_r) e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)}$$

Transmitted fields:

$$\mathbf{E}_t(x, z) = \mathbf{a}_y E_{t0} e^{-j\beta_2(x \sin \theta_t + z \cos \theta_t)}$$

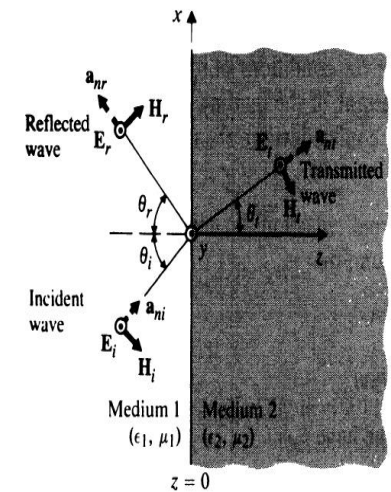
$$\mathbf{H}_t(x, z) = \frac{E_{t0}}{\eta_2} (-\mathbf{a}_x \cos \theta_t + \mathbf{a}_z \sin \theta_t) e^{-j\beta_2(x \sin \theta_t + z \cos \theta_t)}$$

(1) From $E_{1t} = E_{2t}$, $E_{iy}(x, 0) + E_{ry}(x, 0) = E_{ty}(x, 0)$

$$E_{i0}e^{-j\beta_1 x \sin \theta_i} + E_{r0}e^{-j\beta_1 x \sin \theta_r} = E_{t0}e^{-j\beta_2 x \sin \theta_t}$$

(2) From $H_{1t} = H_{2t}$, $H_{ix}(x, 0) + H_{rx}(x, 0) = H_{tx}(x, 0)$

$$\frac{1}{\eta_1} (-E_{i0} \cos \theta_i e^{-j\beta_1 x \sin \theta_i} + E_{r0} \cos \theta_r e^{-j\beta_1 x \sin \theta_r}) = -\frac{E_{t0}}{\eta_2} \cos \theta_t e^{-j\beta_2 x \sin \theta_t}$$



All x should be satisfied, **phase-matching**

$$\beta_1 x \sin \theta_i = \beta_1 x \sin \theta_r = \beta_2 x \sin \theta_t,$$

Reflection law

$$\theta_r = \theta_i$$

Snell's law

$$\sin \theta_r / \sin \theta_i = \beta_1 / \beta_2 = n_1 / n_2$$

$$E_{i0} + E_{r0} = E_{t0}$$

$$\frac{1}{\eta_1} (E_{i0} - E_{r0}) \cos \theta_i = \frac{E_{t0}}{\eta_2} \cos \theta_t$$

Reflection coefficient

$$\Gamma_{\perp} = \frac{E_{r0}}{E_{i0}} = \frac{(\eta_2 / \cos \theta_t) - (\eta_1 / \cos \theta_i)}{(\eta_2 / \cos \theta_t) + (\eta_1 / \cos \theta_i)}$$

$\begin{matrix} h_2 / \cos q_t \rightarrow h_2 \\ \leftarrow \\ h_1 / \cos q_i \rightarrow h_1 \end{matrix}$

Normal incidence

$$\Gamma = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$$

Transmission coefficient

$$\tau_{\perp} = \frac{E_{t0}}{E_{i0}} = \frac{2(\eta_2 / \cos \theta_t)}{(\eta_2 / \cos \theta_t) + (\eta_1 / \cos \theta_i)}$$

$\begin{matrix} h_2 / \cos q_t \rightarrow h_2 \\ \leftarrow \\ h_1 / \cos q_i \rightarrow h_1 \end{matrix}$

$$\tau = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2}{\eta_2 + \eta_1}$$



$$1 + \Gamma_{\perp} = \tau_{\perp}$$

(Note “+” could be “-” in some references)

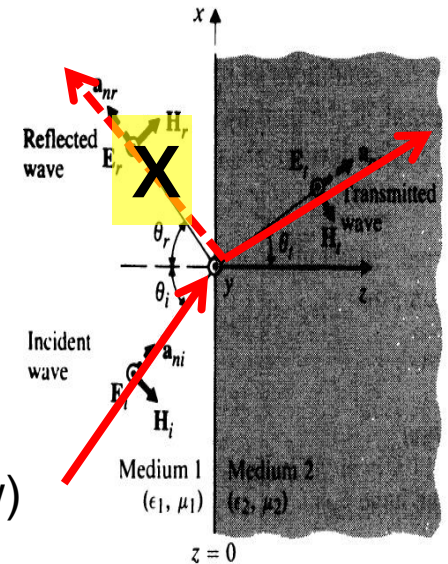
$\Gamma_{\perp} = -1 \quad \tau_{\perp} = 0$ If medium 2 is a perfect conductor,

Brewster angle (布鲁斯特角) for perpendicular polarization

$$\Gamma_{\perp} = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 \cos \theta_i - \eta_1 \cos \theta_t}{\eta_2 \cos \theta_i + \eta_1 \cos \theta_t} = 0 \quad (\text{no reflection})$$

$$\eta_2 \cos \theta_{B\perp} = \eta_1 \cos \theta_i$$

$$\cos \theta_t = \sqrt{1 - \sin^2 \theta_t} = \sqrt{1 - \frac{n_1^2}{n_2^2} \sin^2 \theta_i} \quad (\text{Snell's law})$$



$$\sin^2 \theta_{B\perp} = \frac{1 - \mu_1 \epsilon_2 / \mu_2 \epsilon_1}{1 - (\mu_1 / \mu_2)^2} \rightarrow \mu_1 = \mu_2 = \mu_0, \theta_{B\perp} \text{ does not exist.}$$

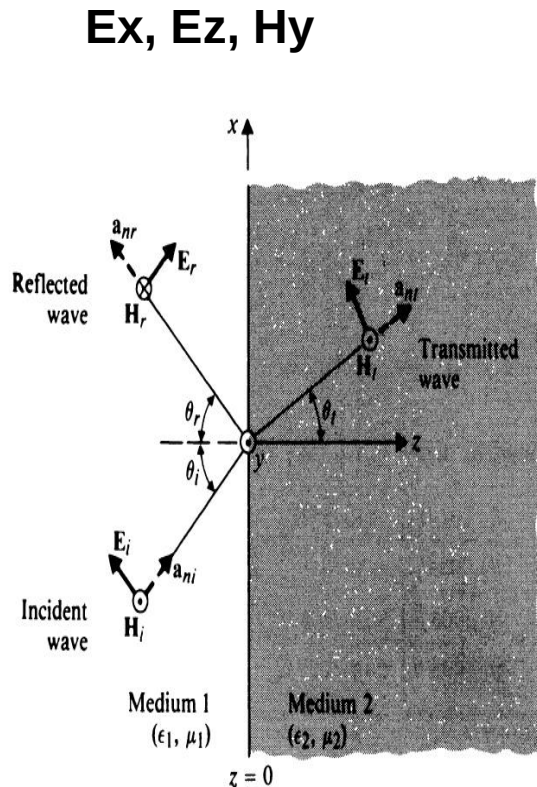
$$\theta_{B\perp} \text{ is called the } \textbf{Brewster angle}, \rightarrow \epsilon_1 = \epsilon_2 \text{ and } \mu_1 \neq \mu_2, \sin \theta_{B\perp} = \frac{1}{\sqrt{1 + (\mu_1 / \mu_2)}}$$

Usually, we don't discuss the Brewster angle for perpendicular polarization.

§ 8-10.3 Parallel polarization (TM, p-polarization)

Fig. 8-21

Parallel polarization (**E** is parallel to the plane of incidence)



Utilizing boundary conditions to solve E_{r0} , θ_r , E_{t0} , θ_t

Incident fields:

$$E_i(x, z) = E_{i0}(\mathbf{a}_x \cos \theta_i - \mathbf{a}_z \sin \theta_i)e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}$$

$$H_i(x, z) = \mathbf{a}_y \frac{E_{i0}}{\eta_1} e^{-j\beta_1(x \sin \theta_i + z \cos \theta_i)}$$

Reflected fields:

$$E_r(x, z) = E_{r0}(\mathbf{a}_x \cos \theta_r + \mathbf{a}_z \sin \theta_r)e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)},$$

$$H_r(x, z) = -\mathbf{a}_y \frac{E_{r0}}{\eta_1} e^{-j\beta_1(x \sin \theta_r - z \cos \theta_r)}.$$

Transmitted fields:

$$E_t(x, z) = E_{t0}(\mathbf{a}_x \cos \theta_t - \mathbf{a}_z \sin \theta_t)e^{-j\beta_2(x \sin \theta_t + z \cos \theta_t)},$$

$$H_t(x, z) = \mathbf{a}_y \frac{E_{t0}}{\eta_2} e^{-j\beta_2(x \sin \theta_t + z \cos \theta_t)}.$$

All x should be satisfied, **phase-matching**

$$(E_{i0} + E_{r0}) \cos \theta_i = E_{t0} \cos \theta_t \quad (8-219)$$

$$\frac{1}{\eta_1} (E_{i0} - E_{r0}) = \frac{1}{\eta_2} E_{t0}. \quad (8-220)$$

Reflection
coefficient:

$$\Gamma_{||} = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}$$

Transmission
coefficient:

$$\tau_{||} = \frac{E_{t0}}{E_{i0}} = \frac{2\eta_2 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i}.$$

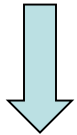
$$\Rightarrow 1 + \Gamma_{||} = \tau_{||} \left(\frac{\cos \theta_t}{\cos \theta_i} \right)$$

Brewster angle (布鲁斯特角) for parallel polarization

$$\Gamma_{||} = \frac{E_{r0}}{E_{i0}} = \frac{\eta_2 \cos \theta_t - \eta_1 \cos \theta_i}{\eta_2 \cos \theta_t + \eta_1 \cos \theta_i} = 0, \text{ (no reflection)}$$



$$\eta_2 \cos \theta_t = \eta_1 \cos \theta_{B||}$$



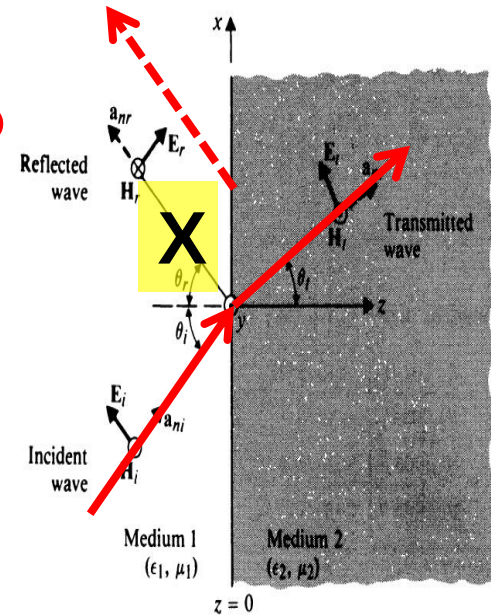
$$\cos \theta_t = \sqrt{1 - \sin^2 \theta_t} = \sqrt{1 - \frac{n_1^2}{n_2^2} \sin^2 \theta_i} \quad (\text{Snell's law})$$

$$\sin^2 \theta_{B||} = \frac{1 - \mu_2 \epsilon_1 / \mu_1 \epsilon_2}{1 - (\epsilon_1 / \epsilon_2)^2}$$

$\theta_{B||}$ is known as the **Brewster angle**

Thus if $\mu_1 = \mu_2$,

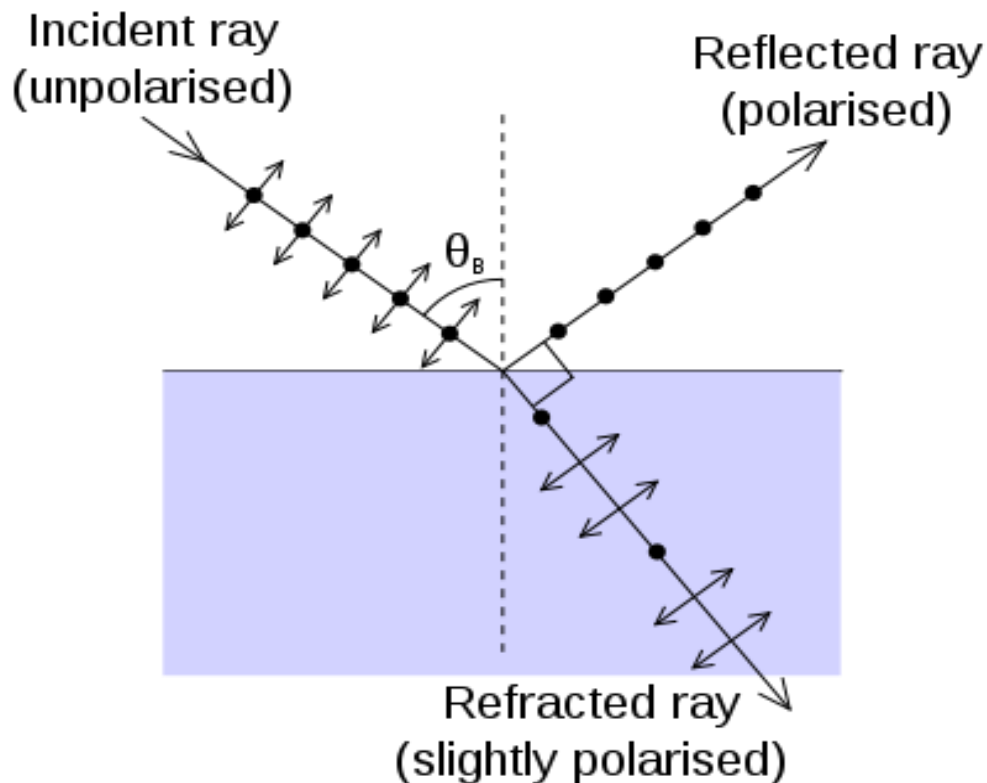
$$\theta_{B||} = \tan^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \tan^{-1} \left(\frac{n_2}{n_1} \right)$$



An application of the Brewster angle

Since the formula of the Brewster angles for perpendicular and parallel polarizations are different, it is possible to separate these two types of polarizations in an unpolarized wave.

For example, the perpendicularly polarized wave is reflected, while the parallelly polarized wave is refracted.



§ 8-10.1 Total reflection (全反射)

$$n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$n_1 = \sqrt{\epsilon_1}, \quad n_2 = \sqrt{\epsilon_2}$$

Critical angle ($\theta_2=90^\circ$):

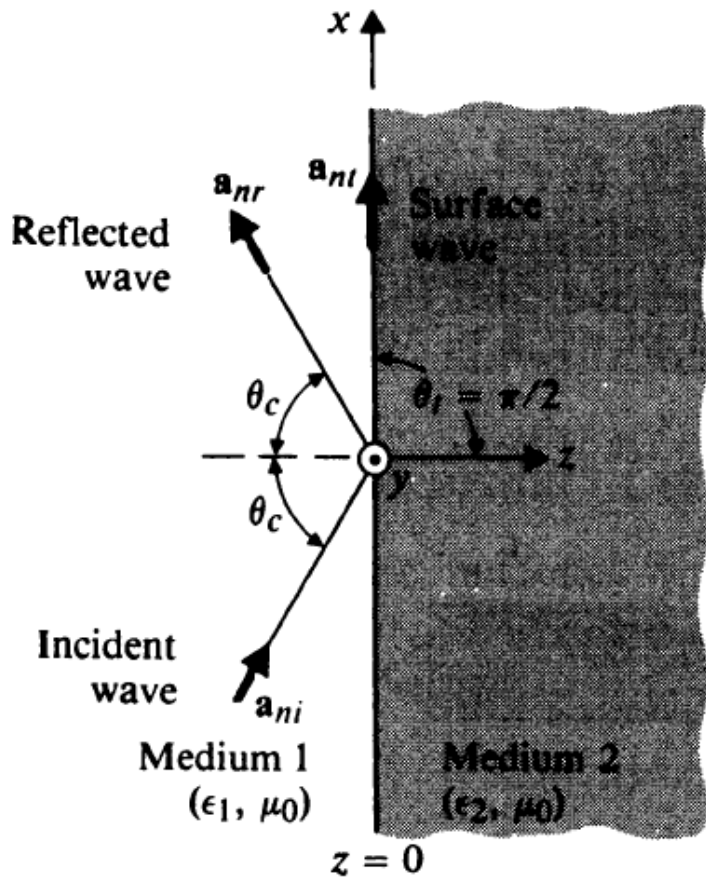
$$\theta_c = \sin^{-1} \sqrt{\frac{\epsilon_2}{\epsilon_1}} = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

Wave in medium 2 (倏逝波):

$$\sin \theta_t = \sqrt{\frac{\epsilon_1}{\epsilon_2}} \sin \theta_i > 1,$$

$$\cos \theta_t = \sqrt{1 - \sin^2 \theta_t} = \pm j \sqrt{\frac{\epsilon_1}{\epsilon_2} \sin^2 \theta_i - 1}.$$

$$e^{-j\beta_2 \mathbf{a}_{nt} \cdot \mathbf{R}} = e^{-j\beta_2 (x \sin \theta_t + z \cos \theta_t)},$$



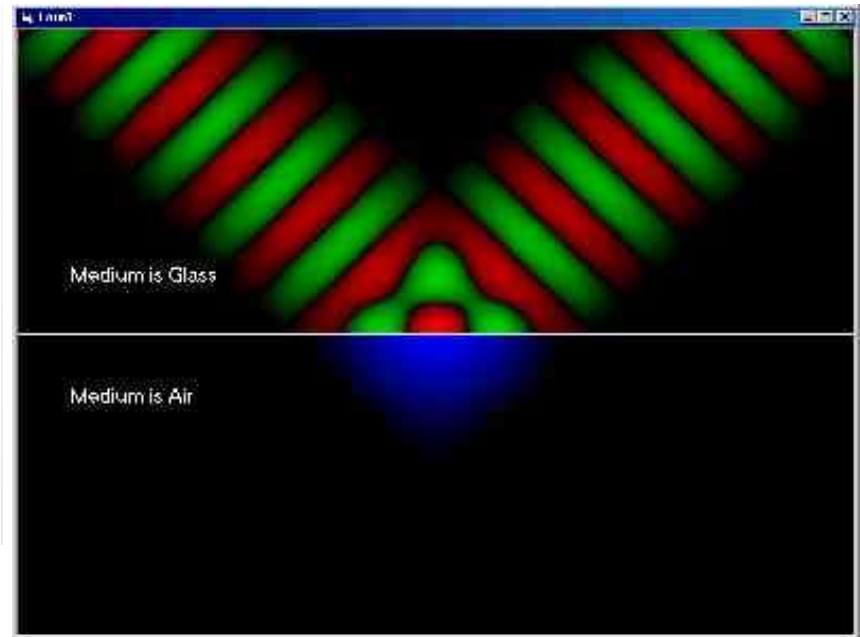
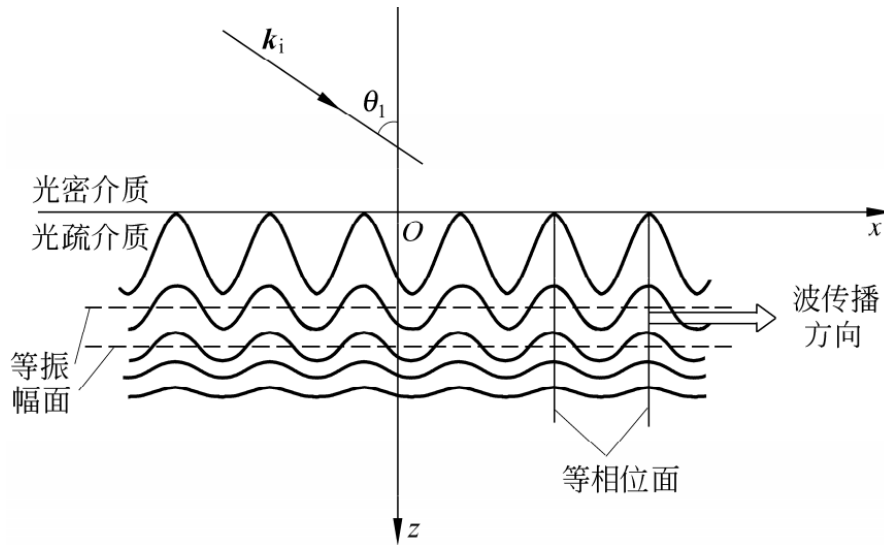
Plane wave incident at critical angle, $\epsilon_1 > \epsilon_2$

Evanescent decay (倏逝衰减) along z axis
but propagating along the surface.

Evanescent Wave (倏逝波)

Transmitted field: $E_t = A_t \exp(\mp \kappa z) \exp[i(k_{tx} x - \omega t)]$

Propagate: x-direction; Attenuate: z-direction



8-15 The dielectric constant of pure water is 80

- (a) Determine the Brewster angle for parallel polarization and corresponding angle of transmission**
- (b) A plane wave with perpendicular polarization is incident from air on water surface at Brewster angle. Find the reflection and transmission coefficients**